Emily Fang

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EDUCATION

North Carolina State University, Raleigh, NC

Ph.D. in Industrial Engineering (GPA: 3.94), Human Factors Engineering and Human-Computer Interaction — *Aug 2024 – Present*

B.S. in Industrial Engineering (GPA: 3.77), Minor in Supply Chain Management — *Aug 2019 – May 2024*

RESEARCH EXPERIENCE

North Carolina State University, Graduate Research Assistant — *May 2024 – Present*

Astrophysics at Scale: Transforming and Evaluating Undergraduate Education through Virtual and Desktop Environments *Oct 2026 – Present*

- Completed training in the Reformed Teaching Observation Protocol (RTOP) to conduct classroom observations for Astroverse, an NSF-funded project (Award #2540028) and the next version of Scale Worlds.

AI-Assisted Augmented Reality Checklists for Occupational Hazard Recognition and Mitigation *(GitHub)* — *Aug 2025 – Oct 2026*

- Developed an augmented reality prototype in Unity (C#), deploying to Meta Quest 3 using the passthrough camera API to overlay hazard-mitigation checklists in the user's real environment.
- Integrated OpenAI GPT-4o via API with a gesture-driven Unity UI, sending Meta Quest 3 passthrough camera snapshots to generate context-specific hazard-mitigation checklists and parsing structured JSON into interactive canvas elements.
- Evaluated 10 prompt versions against 42 expert-annotated construction images using an LLM judge (GPT-4.1). The deployed prompt improved precision from 0.63 to 0.75 while maintaining recall (0.70 to 0.72).

Short-Term Cognitive Effects of Different Video Game Tasks — *May 2024 – Present*

- Designed and administered 2-hour in-lab experimental protocols with adult participants (n = 46, three between-subjects conditions), conducting standardized cognitive assessments to measure selective attention, inhibitory control, working memory, and multitasking abilities pre- and post-intervention.
- Facilitated computer-based tasks, conducted semi-structured interviews, and collected behavioral performance, eye-tracking, and survey data (Qualtrics, PANAS, NASA-TLX) for quantitative and qualitative analysis.
- Analyzed data using ANCOVA in R and thematic analysis. Found that FPS play increased self-reported temporal demand, effort, and negative affect, while cognitive performance did not differ across conditions.

Scale Cognition Advanced Learning Environments in Virtual Reality (SCALE-VR) — *Aug 2023 – Jul 2025*

- Collaborated with interdisciplinary teams across education, design, and engineering (led by 3 PIs) to assist in the iterative development of an NSF-funded VR application (Award #2055680) designed to teach students concepts of scale and numeracy, presenting findings at weekly team meetings.
- Conducted all qualitative coding and thematic analysis of formative evaluation data (30 think-aloud sessions and 15 semi-structured interviews across HMD and CAVE versions), identifying insights to enhance usability and user experience that informed three design changes.
- Coded bipolar laddering responses from A/B preference testing with 10 usability experts.
- Trained an undergraduate researcher in thematic analysis over a semester; he subsequently coded qualitative data independently.

North Carolina State University, Undergraduate Research Assistant — *Aug 2021 – May 2024*

Augmented Reality for Engineering Education Enhancement (AREEA) — *Oct 2022 – May 2024*

- Collaborated with a Ph.D. researcher across all phases of a semester-long, mixed-methods evaluation (n = 42) of a game-based augmented reality application (HoloLens 2) aimed at teaching statistics, engaging directly with participants to collect both qualitative and quantitative data.
- Facilitated data collection by administering validated questionnaires, conducting interviews, and guiding participants through evaluation tasks. Conducted qualitative and quantitative analysis to generate insights on usability and UX.
- Recorded task completion times and misclick counts by casting the HoloLens 2 display to a PC, took session notes, graded content assessments, and coded interview data.

Virtual Instructor Application 2 (VIA 2) — *Aug 2021 – Oct 2022*

- Assisted a Ph.D. researcher to iteratively improve an augmented reality-based postural training tool for occupational safety, applying both qualitative and quantitative research methods to evaluate and refine usability and effectiveness.
- Utilized various research methodologies, including PSSUQ (Post-Study System Usability Questionnaire), Bipolar Laddering, and Semi-Structured Interviews to deliver actionable insights on tool usability and training outcomes.
- Conducted thematic analysis as one of two independent coders (NVivo) of data from 13 manual material handling workers, identifying issues with element salience and gesture responsiveness.

PUBLICATIONS

Cheng, F., **Fang, E.**, & Chen, K. B. (2026). Elicitation and assessment of hand-based gestures for information processing in immersive environments. *Human Factors: The Journal of the Human Factors and Ergonomics Society*.

Fang, E., & Chen, K. B. (2026). Examining short-term video game play on subsequent working memory performance. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

Chen, K. B., Alabi, C., Bailey, S. K. T., **Fang, E.**, Fussell, S. G., Harris, J., Akash, K., & Wu, L. (2026). Human Factors Extended Reality Showcase 2026. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

Perera, G. N., **Fang, E.**, Bottomley, L., Chen, K. B., & Ivy, J. (2025). Study design and assessment framework for testing augmented reality tools in engineering education. *2025 ASEE Annual Conference & Exposition*. **1st Place, Best New Career Paper Award.**

Fang, E., Kulasingam, R., Cheng, F., Peterson, M., Delgado, C., & Chen, K. B. (2025). Examining user interactions with signaling elements in a virtual reality learning application. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

Cheng, F., **Fang, E.**, & Chen, K. B. (2025). Mental models of gestural interaction for information processing in virtual reality. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

Fang, E., Sivaramakrishnan, A., & Chen, K. B. (2024). Negative emotions from virtual reality usage: A preliminary exploratory study using online forums. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

Chen, K., Perera, G., **Fang, E.**, & Chen, K. B. (2022). Augmented reality (AR) posture training for manual material handling: Iterative design, evaluation, and recommendation for AR interface. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

Perera, G. N., **Fang, E.**, Chen, K., Ivy, J. S., & Chen, K. B. (under review). Worker-centered design and evaluation of augmented reality application for posture training. *Ergonomics in Design*.

Wu, L., **Fang, E.**, Cheng, F., & Chen, K. B. (in preparation). Balancing usability and theory: An iterative evaluation framework for virtual learning environments. Target journal: *International Journal of Human–Computer Interaction*.

PRESENTATIONS AND DEMONSTRATIONS

Augmented Reality Hazard Detection: An AI-Assisted Approach to Construction Safety. XR Showcase demo (3 hr), *HFES Annual Meeting*, 2026. Co-presented with M. Malaek and M. Roy.

Examining Short-Term Video Game Play on Subsequent Working Memory Performance. Poster presentation, *HFES Annual Meeting*, 2026.

Human Factors Extended Reality Showcase 2025: Scale Worlds. XR Showcase demo (3 hr), *HFES Annual Meeting*, 2025.

Examining User Interactions With Signaling Elements in a Virtual Reality Learning Application. Lecture presentation (20 min), *HFES Annual Meeting*, 2025.

Negative Emotions From Virtual Reality Usage: A Preliminary Exploratory Study Using Online Forums. Lecture presentation (20 min), *HFES Annual Meeting*, 2024.

Stratify Subjective Numeric Ratings and Qualitative Feedback of Augmented Reality Ergonomic Posture Training Tool. Poster presentation, *IISE Applied Ergonomics Conference*, 2023.

HONORS AND AWARDS

National Institute of Occupational Safety and Health Graduate Fellowship at NC State University, *2025 – Present*

Provost's Doctoral Fellowship at NC State University, *2024 – 2025*

Graduate Merit Award at NC State University, *2024 – 2025*

Institute of Industrial and Systems Engineers Applied Ergonomics Conference Student of the Year Award, *2023*

TEACHING AND PROFESSIONAL SERVICE

Grader, ISE 540: Human Factors in Systems Design, North Carolina State University, Spring 2026

Reviewer, *HFES Annual Meeting*, 2025 – 2026

Session Chair, XR6: Understanding Human Performance and Physiological Responses in XR, Extended Reality Technical Group, *HFES Annual Meeting*, 2025

OUTREACH

- Led hands-on demonstrations of Scale Worlds, an NSF-funded VR learning environment, at the NC Science Museum and summer camps; assisted with demonstrations at a local underserved middle school, NC State Engineering Open House, and lab tours for visiting seminar speakers
- Counselor, Industrial and Systems Engineering Summer Camp for High School Students, The Engineering Place, North Carolina State University, Summers 2025 – 2026. Mentored about 25 high school campers on college, research, and ISE; designed interactive end-of-camp review games covering manufacturing and human factors tools.

SKILLS

Programming & XR: Python, PyTorch, scikit-learn, C#, JavaScript, R, Unity3D, Meta XR SDK, OpenAI API, Streamlit, Gurobi, Git/GitHub
Research software: Qualtrics, Dedoose, NVivo, Tableau, Figma, PsyToolkit, Pupil Labs (Pupil Core)
Methods: Mixed Methods, Qualitative and Quantitative Analysis, Signal Detection Theory, Experimental Design, Eye Tracking & Pupillometry, Thematic Analysis, Usability Analysis, Formative Evaluation, Think-Aloud Protocols, User-Centered Design, Study Coordination, Technical Writing
Experimental paradigms: n-back, flanker, task switching | **Languages:** English, Mandarin (native/bilingual)
Professional memberships: Institute of Industrial and Systems Engineers (IISE), 2022 – Present; Human Factors and Ergonomics Society (HFES), 2022 – Present